SEMINAR SUMMARY: MARCH 31, 2026 - NEAL BUSHAW BOOTSTRAP PERCOLATION ON GRIDS.

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Percolation theory is a branch of mathematics rooted in time based graphical shading changes related to a set of rules and an initial condition. We were presented with an $n \times m$ square tiling of the plane. Our initial condition was a collection of shaded blocks, take Figure 1 for example. The percolating mechanic was then set to being that if a square has k shaded (or infected) neighbors then in the next time step, it will become shaded, and time steps in definite intervals. For example, if $k = 1$, the behaviour is as if each square becomes a "source" in the sense of water or sand, and over time the entire board is filled so long as there is a single shaded block at our initial condition.

We then began looking at the scenario for $k = 2$ (i.e. two neighbors, as edge neighbors, not corner neighbors). If we were to take Figure 1, and step it once in time, then we would get Figure 2.

Eventually this initial value percolation from Figures 1 and 2 fills the entire space except for the right column of squares. Oddly some of these bootstrap percolations fill the entire space, some leave just edges behind, and some stop entirely. The goal behind the research here is to classify the different categories of initial conditions.

We then explored a considerable number of different bootstraps (or initial conditions) randomdomly generated for the $k = 2$ problem. It was interesting to see which percolated to

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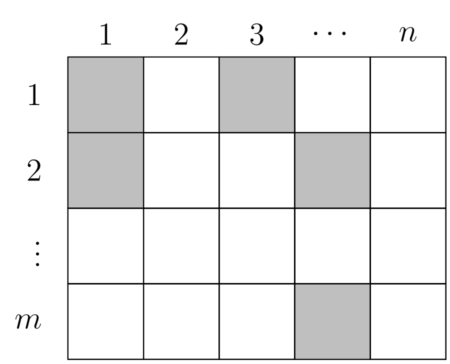

$G_{4,5}$

FIGURE 1. A rectangular $G_{4,5}$ tiled with squares, five shaded as bootstrap for percolation.

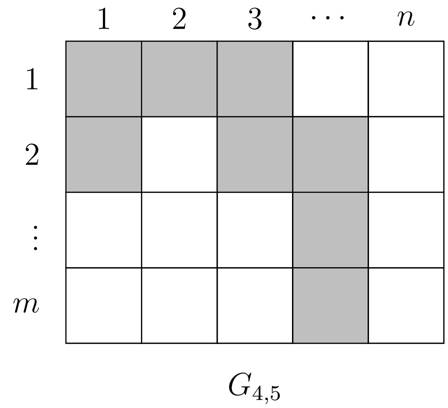

FIGURE 2. A rectangular $G_{4,5}$ tiled with squares, eight shaded after time step 1 with $k = 2$.

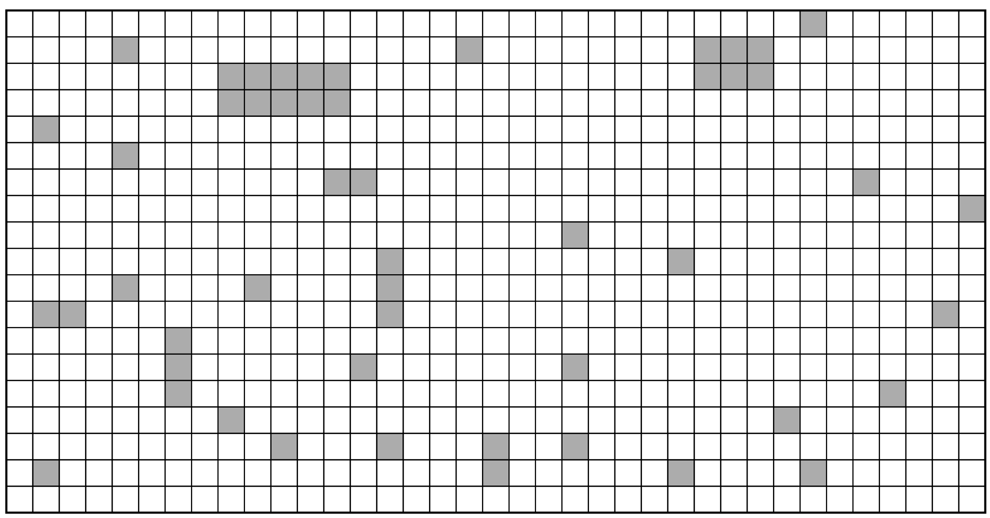

FIGURE 3. A large square based Bootstrap Percolation with $k = 2$ in a steady state

the entire domain and which stopped. Figure 3 gives us this steady state where percolation cannot continue. A question was then asked: what is the minimal number of squares needed filled such that a $k = 2$ completely percolates to the entire domain. The answer curiously was that it need be a single diagonal from a corner to the other corner, for the square grid case. Take for example the identity matrix as an example.

We then switched our attention to the square bootstrap percolation with $k = 3$, and we found it much more difficult to get a percolation to the entire domain as there were far more stable configurations of squares. Furthermore, there was some subtle research going on about $k = 3$ as recently as 2023 and 2025.

Finally, we were presented with a variety of general tools for approaching the problem.

Generally the problem seemed to garner quite a good amount of attention and engagement from the class which was fun. The problem itself seemed intuitive to follow at first, and then when k was varied intuition was completely lost, making the problem different and interesting. A question was asked about applicability, and Professor Bushaw said that he was not concerned with application merely the curiosity of the problem which was interesting in its own right. It was definitely a problem that I could find myself exploring because I have a background in tilings over infinite words on the n letter alphabet, but I do not know if that outweighs my interest in pure logic.